

Stata Group Assignment 1

Research Methods in Finance | Olin Business School

Operating-profitability decile portfolios

1. Data and portfolio selection

The analysis uses the ten **value-weighted** portfolios formed on operating profitability (OP) from Kenneth French's Data Library. Decile 1 contains the lowest-profitability stocks and decile 10 the highest. Portfolios are formed each June using NYSE breakpoints. OP is prior-year revenue minus cost of goods sold, interest expense, and selling, general and administrative expenses, divided by book equity.

Monthly portfolio returns are matched by calendar month to RF from the Fama/French three-factor file. The common sample is **July 1963-July 2026 (757 months)**. Files were downloaded on September 15, 2026 and use the July 2026 CRSP database. All ten deciles and RF have complete observations in this sample. Only the monthly value-weighted return block is used.

2. Excess returns and tests

For decile d in month t , excess return = portfolio return - RF. Both inputs are in percent per month, so an excess return of 0.69 denotes 0.69 percentage points. For each decile, the two-sided null is H_0 : mean excess return = 0. The statistic is $t = \text{mean} / (s / \sqrt{N})$, where s is the sample standard deviation ($N - 1$ denominator). The p-value uses a Student t distribution with 756 degrees of freedom.

3. Summary statistics and hypothesis tests

OP decile	N	Mean (%)	SD (%)	t statistic	p-value
1 (low)	757	0.4394	6.5959	1.8327	0.0672363
2	757	0.5142	5.3723	2.6336	0.0086206
3	757	0.4696	4.9263	2.6228	0.0088974
4	757	0.5619	4.6365	3.3345	0.0008961
5	757	0.6659	4.6271	3.9593	0.0000823
6	757	0.5924	4.5478	3.5837	0.0003605
7	757	0.5573	4.5699	3.3554	0.0008320
8	757	0.7272	4.6130	4.3375	0.0000164
9	757	0.7089	4.5306	4.3048	0.0000189
10 (high)	757	0.6910	4.6210	4.1143	0.0000431

Notes: Monthly arithmetic means and sample standard deviations; two-sided, unadjusted p-values. Calculations use Stata's one-sample ttest command. Source: Kenneth French Data Library.

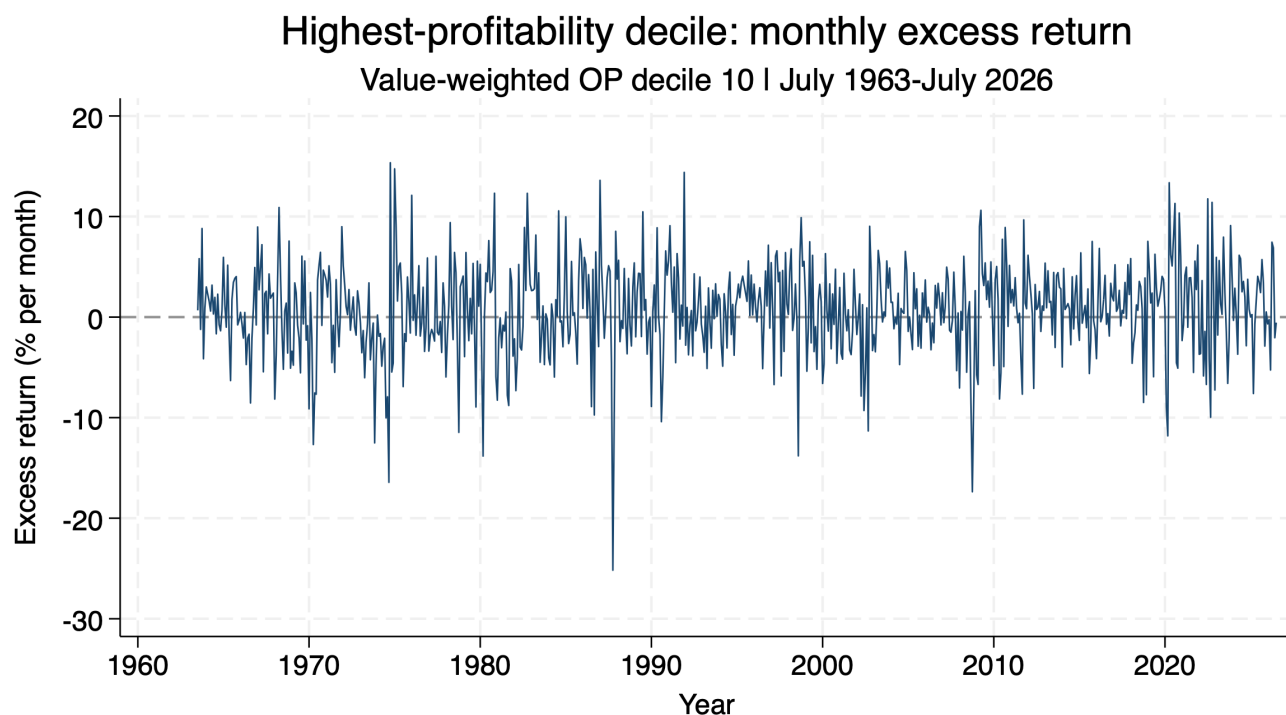
Discussion and time-series figure

All ten deciles have positive sample mean excess returns, ranging from 0.4394% to 0.7272% per month. At the 5% significance level, the zero-mean null is rejected for deciles 2-10. Decile 1 has $t = 1.8327$ and $p = 0.0672$, so it is not significant at 5%, although it is significant at 10%. Deciles 2-10 are also individually significant at 1%.

Higher-profitability portfolios generally have larger mean excess returns, but the pattern is not monotonic: decile 8 has the highest mean (0.7272%), while decile 10 averages 0.6910%. Decile 1 is the most volatile ($SD = 6.5959\%$), compared with 4.6210% for decile 10. The decile-10 minus decile-1 difference in sample means is about 0.252 percentage points per month. These individual tests do not establish that the difference itself is statistically significant.

These are conventional one-sample t-tests. Their standard errors do not adjust for serial correlation, and the ten p-values are not corrected for multiple testing. Positive excess returns do not establish risk-adjusted abnormal performance or a causal effect of profitability; no asset-pricing factor adjustment is made.

4. Highest-profitability decile over time



Source: Kenneth French Data Library; portfolio return minus RF.

Figure 1. Decile 10 monthly excess returns fluctuate substantially around zero despite a positive sample mean. The largest negative observation is October 1987 (-25.17%). The plot shows monthly returns, not cumulative investment performance.

Sources and reproducibility

[Kenneth French Data Library](#); [operating-profitability portfolio definitions](#). RF uses Ibbotson one-month Treasury bill data through May 2024 and the ICE BofA US 1-Month Treasury Bill Index thereafter, per the downloaded file. The library uses its current CIZ-based research return series.

The week-02 folder preserves the downloaded ZIP/CSV files, runnable assignment1.do, merged dataset, results table and graph. Set the Stata working directory to the repository root and run run-week-02.do. A local log is generated on each run. Row ranges refer to the preserved July 2026 files.

Appendix: Stata code

```
version 19.0
clear all
set more off
* Run from week-02, or use run-week-02.do from the repository root.
* Frozen July 2026 data vintage.
capture mkdir output
capture log close assignment1
log using "output/assignment1.log", text replace name(assignment1)

* 1. Downloaded from Kenneth French's Data Library on 15 September 2026.
* Raw ZIPs and CSVs are preserved in data/raw for reproducibility.
* Original URLs:
* https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Portfolios_Formed_on_OP_CSV.zip
* https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_Factors_CSV.zip
* Row ranges select only the monthly blocks in this frozen vintage.
* OP columns 10-19 are the ten value-weighted decile portfolios.
import delimited using "data/raw/Portfolios_Formed_on_OP.csv", ///
    rowrange(26:782) varnames(nonames) clear
rename v1 yyyymm
forvalues d = 1/10 {
    local c = `d' + 9
    rename v`c' ret`d'
}
keep yyyymm ret1-ret10
isid yyyymm
tempfile portfolios
save `portfolios'

* Monthly RF is column 5 of the Fama/French three-factor file.
import delimited using "data/raw/F-F_Research_Data_Factors.csv", ///
    rowrange(6:1206) varnames(nonames) clear
rename v1 yyyymm
rename v5 rf
keep yyyymm rf
isid yyyymm
merge 1:1 yyyymm using `portfolios'
assert _merge != 2
keep if _merge == 3
drop _merge
generate month = ym(floor(yyyymm/100), mod(yyyymm,100))
format month %tm
sort month
tsset month
assert month == month[_n-1] + 1 if _n > 1
assert _N == 757
assert month[1] == ym(1963,7) & month[_N] == ym(2026,7)

* 2. Returns and RF are in percent per month; retain these units.
foreach v of varlist ret1-ret10 rf {
    replace `v' = . if inlist(`v', -99.99, -999)
    assert !missing(`v')
}
forvalues d = 1/10 {
    generate double excess`d' = ret`d' - rf
    label variable excess`d' "OP decile `d': excess return (%)"
}
label variable month "Month"
label variable rf "One-month risk-free return (%)"
label data "Value-weighted OP deciles: July 1963-July 2026"
save "output/op_excess_returns.dta", replace
```

Appendix: Stata code (continued)

```
export delimited using "output/op_excess_returns.csv", replace

* 3. One-sample, two-sided t tests: H0: E[excess_d] = 0.
* t = mean / (sample SD / sqrt(N)); p = 2*ttail(N-1,abs(t)).
tempfile statistics
tempname results
postfile `results' byte decile int n double mean sd t p ///
    using `statistics', replace
forvalues d = 1/10 {
    quietly ttest excess`d' == 0
    post `results' (`d') (r(N_1)) (r(mu_1)) (r(sd_1)) (r(t)) (r(p))
}
postclose `results'
preserve
use `statistics', clear
format mean sd t %9.4f
format p %10.7f
list, noobs separator(0)
save "output/summary_statistics.dta", replace
export delimited using "output/summary_statistics.csv", replace
restore

* 4. Time-series figure: highest operating-profitability decile.
twoway line excess10 month, ///
    title("Highest-profitability decile: monthly excess return") ///
    subtitle("Value-weighted OP decile 10 | July 1963-July 2026") ///
    ytitle("Excess return (% per month)") xtitle("Year") ///
    xlabel(0(120)720, format(%tmCCYY)) ///
    yline(0, lcolor(gs9) lpattern(dash)) ///
    lcolor(navy) lwidth(thin) graphregion(color(white)) ///
    note("Source: Kenneth French Data Library; portfolio return minus RF.") ///
    name(op_decile10, replace)
graph save "output/op_decile10.gph", replace
graph export "output/op_decile10.png", width(2400) replace
log close assignment1
```